ADC Lab Logbook

Part 1 | Section 3: Resonance and Filters

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Before the Lab: Second-Order Low-Pass Filter Design

The target specification for group a21 was to design a second-order low-pass RLC filter with a corner frequency of $f_c = 7400\ Hz$. The transfer function of the series RLC low-pass filter is:

$$V_{out} / V_{in} = (1/j\omega C) / (1/j\omega C + j\omega L + R) = 1 / ((j\omega)^2 LC + j\omega RC + 1)$$

The corner (natural) frequency is given by $\omega_c = 1/\sqrt{(LC)}$. Selecting C = 100 nF and solving for L:

$$(2\pi \times 7400)^2 = 1/(LC) \implies L = 1/((2\pi \times 7400)^2 \times 10^{-7}) \approx 4.63\ mH \approx 4.7\ mH\ (nearest\ standard)$$

For a critically-damped response ($\zeta = 1$), the damping condition requires $R^2C^2 = 4LC$, giving:

$$R = 2\sqrt{(L/C)} = 2\sqrt{(4.7 \times 10^{-3} / 10^{-7})} \approx 433\ \Omega$$

The LTspice circuit was constructed with R = 433 Ω, L = 4.7 mH, C = 100 nF, and an AC sweep was configured from 10 Hz to 100 kHz with 100 points per decade to simulate the ideal frequency response.

Resonant LC Network

At resonance, the inductive and capacitive reactances are equal and opposite, so the total reactive impedance is zero: $j\omega L + 1/(j\omega C) = 0$, giving $\omega = 1/\sqrt{(LC)}$. Using L = 100 mH and C = 100 nF:

$$\omega = \sqrt{(1/(10^{-1} \times 10^{-7}))} = 10^4\ rad/s \implies f = 10^4 / (2\pi) \approx 1591\ Hz$$

The total impedance of the series RLC network is $Z = R + j(\omega L - 1/\omega C)$. Resonance corresponds to the condition under which the imaginary part vanishes and |Z| is at its minimum.

RLC in Series

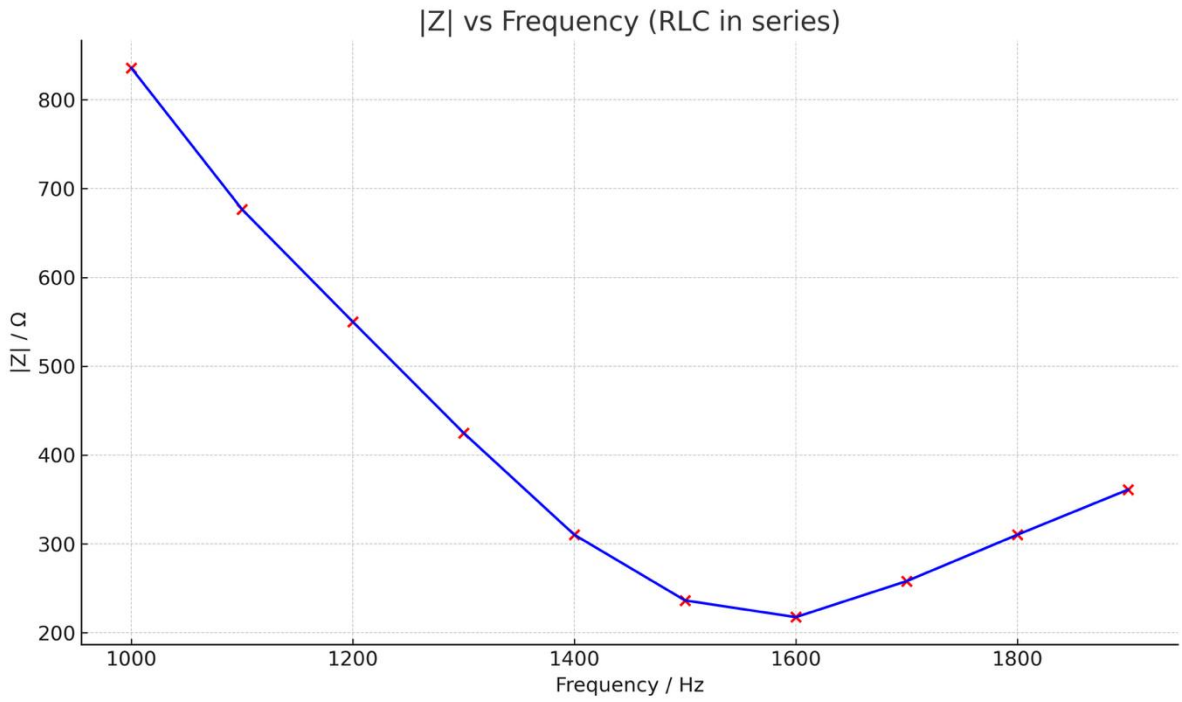
The impedance of the series RLC combination (L = 100 mH, C = 100 nF) was measured using the standard potential divider method with R = 100 Ω, sweeping from 1 kHz to 1.9 kHz.

Frequency (Hz)	R (Ω)	Vz (mV)	Vr (mV)	|Z| (Ω)
1k	100	468	56	835.7
1.1k	100	460	68	676.5
1.2k	100	440	80	550
1.3k	100	408	96	425
1.4k	100	360	116	310.3
1.5k	100	312	132	236.4
1.6k	100	296	136	217.6
1.7k	100	320	124	258.1
1.8k	100	360	116	310.3
1.9k	100	396	108	361.1

All results are based on experimentally measured data, not numerical or circuit simulations.

The measured impedance magnitude exhibits a clear minimum between 1.5 kHz and 1.7 kHz,

consistent with the theoretical resonant frequency of 1591 Hz. At resonance in a series circuit, the inductive and capacitive reactances cancel and |Z| is reduced to the parasitic resistance of the components.



RLC in Parallel

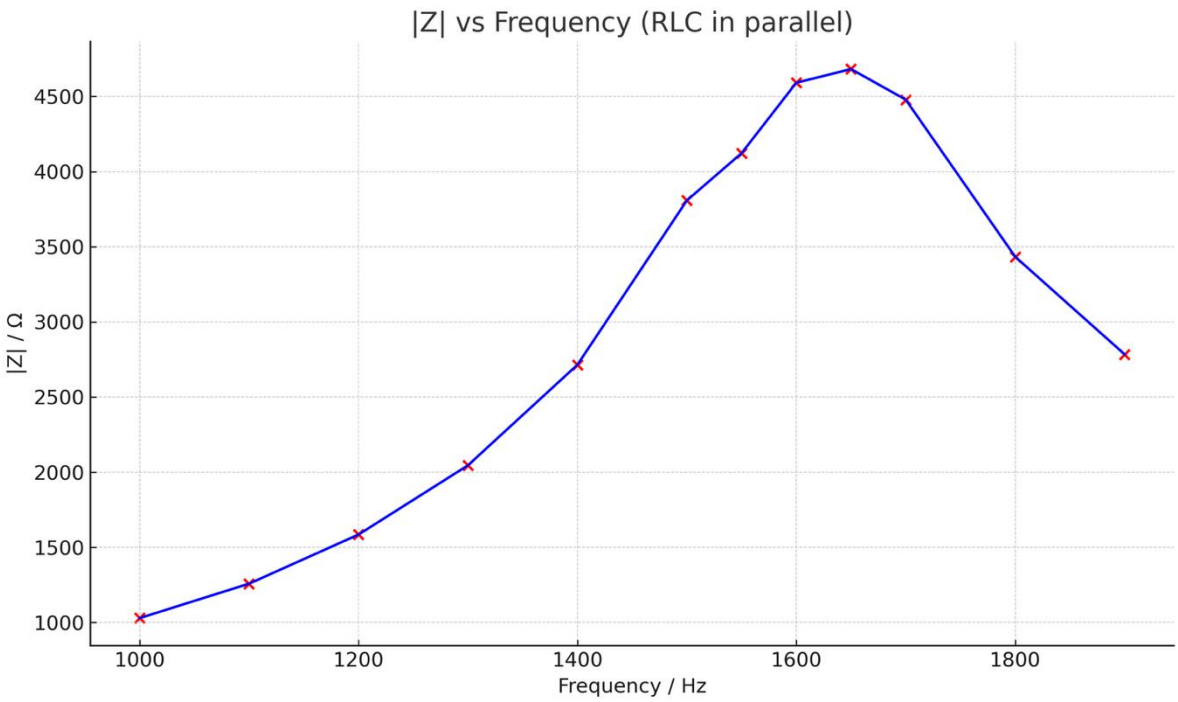
The parallel RLC combination was measured over the same frequency range using R = 1 kΩ. Initial measurements at 100 Hz intervals showed a broad peak; additional points were taken at 1.55 kHz and 1.65 kHz to improve resolution.

Frequency (Hz)	R (Ω)	Vz (mV)	Vr (mV)	|Z| (Ω)
1k	1k	284	276	1029
1.2k	1k	336	212	1584.9
1.4k	1k	380	140	2714.3
1.5k	1k	396	104	3807.7
1.55k	1k	404	98	4122.4
1.6k	1k	404	88	4591
1.65k	1k	412	88	4681.8
1.7k	1k	412	92	4478.3
1.8k	1k	412	130	3433.3
1.9k	1k	412	148	2783.8

All results are based on experimentally measured data, not numerical or circuit simulations.

The parallel circuit exhibits an impedance **maximum** near resonance (approximately 1650 Hz), in contrast to the series case. At resonance in a parallel circuit, the circulating current between L and C is large, but the net current drawn from the source is minimal, resulting in a high effective impedance.

The graph is in the next page



Discussion: Non-Ideal Resonance Behaviour

In an ideal series RLC circuit, the LC impedance would be exactly zero at resonance; in an ideal parallel RLC circuit, it would be infinite. In practice, neither condition is achieved. The measured series minimum and parallel maximum are finite because the imaginary parts of the inductor and capacitor impedances cancel, but their real parts (parasitic resistances) do not. Since both components are wound or fabricated from metallic conductors, a residual resistive component is always present, limiting the sharpness of the resonance (i.e. reducing the Q-factor).

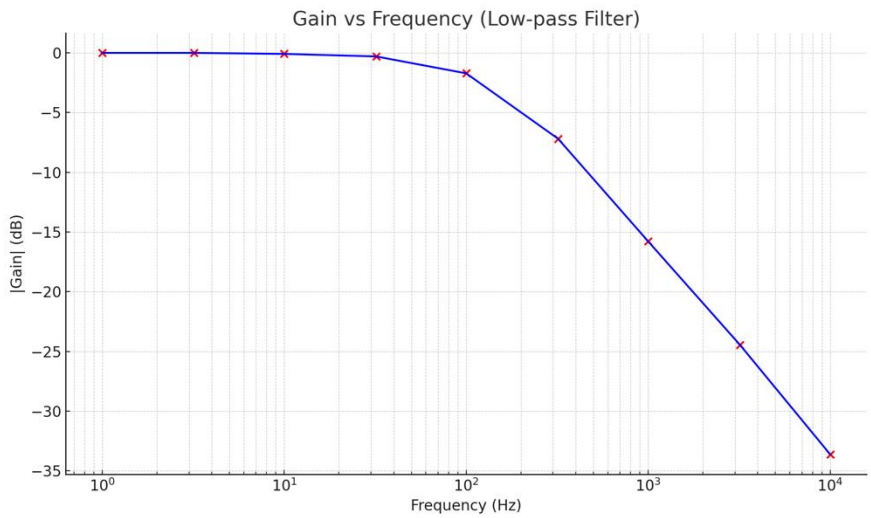
First-Order Low-Pass Filter

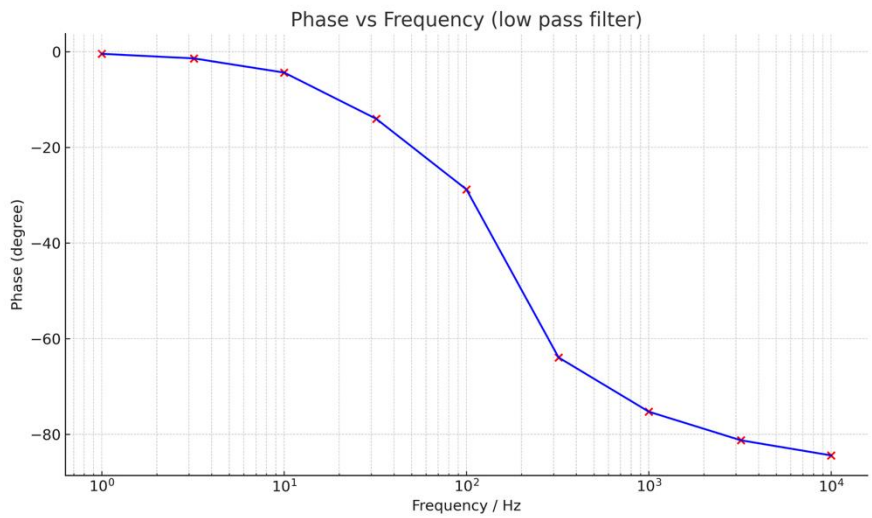
A first-order RC low-pass filter was constructed with R = 1 kΩ and C = 1 μF (theoretical −3 dB frequency: $f_c = 1/(2\pi RC) \approx 159$ Hz). The gain and phase were measured from 1 Hz to 10 kHz.

Frequency (Hz)	Vz (mV)	Vin (mV)	Gain	Phase (°)
1	1010	1010	1.000	-0.44
3.2	1010	1010	1.000	-1.4
10	1000	1010	0.990	-4.36
32	976	1010	0.966	-14.03
100	808	984	0.821	-28.8
320	416	952	0.437	-63.92
1000	156	960	0.163	-75.25
3200	56	936	0.060	-81.23
10000	19.6	940	0.021	-84.39

All results are based on experimentally measured data, not numerical or circuit simulations.

The Bode plot confirms the expected low-pass response: unity gain at low frequencies, with the gain rolling off at approximately −20 dB/decade above f_c. The phase approaches −90° asymptotically at high frequencies, consistent with theory.





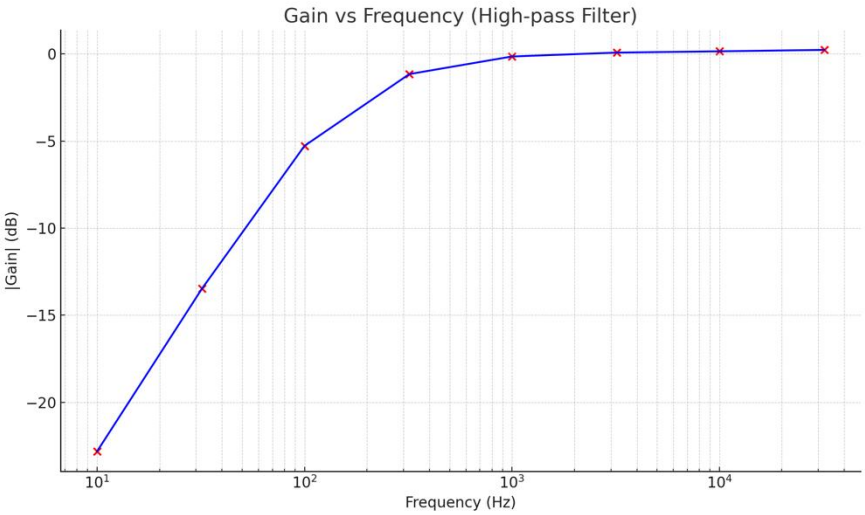
First-Order High-Pass Filter

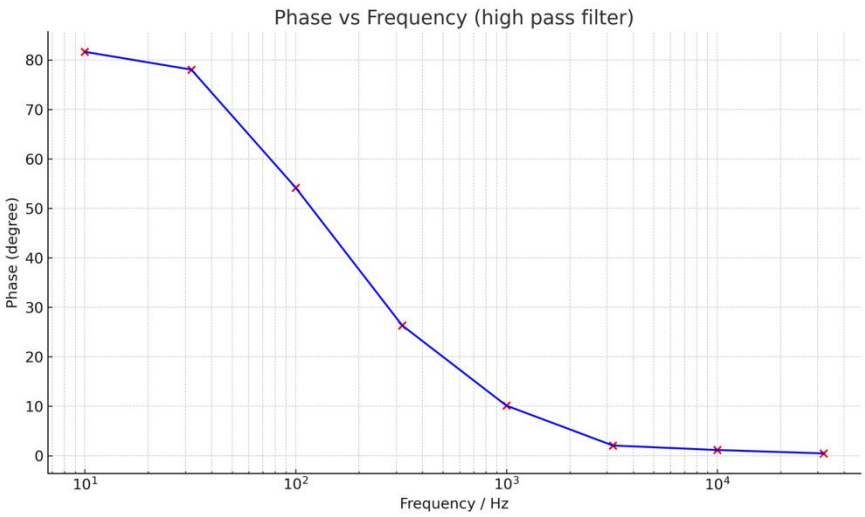
The RC high-pass filter was implemented by swapping the positions of R and C (output taken across R). The same component values were used (R = 1 kΩ, C = 1 μF).

Frequency (Hz)	Vz (mV)	Vin (mV)	Gain	Phase (°)
10	75.2	1040	0.072	81.65
32	216	1020	0.212	78.06
100	536	984	0.545	54.2
320	840	960	0.875	26.35
1000	928	944	0.983	10.12
3200	944	936	1.008	2.06
10000	952	936	1.017	1.15
32000	920	896	1.027	0.46

All results are based on experimentally measured data, not numerical or circuit simulations.

The high-pass filter attenuates low-frequency signals and passes high-frequency signals, as expected. The gain rolls off below f_c at −20 dB/decade, and the phase leads by approximately +90° at low frequencies, converging to 0° at high frequencies.



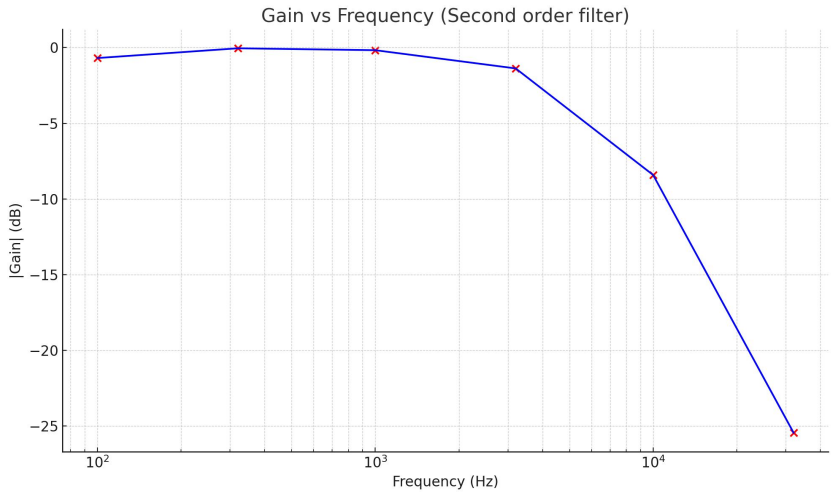
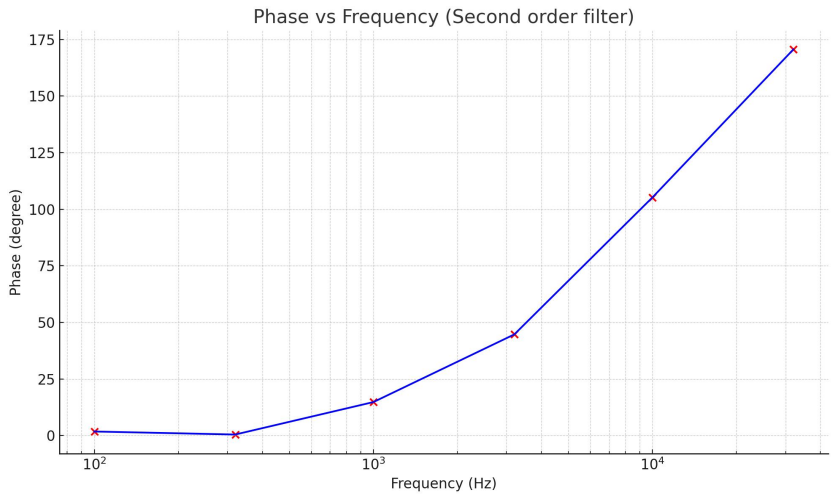


Challenge: Second-Order Low-Pass Filter (Group a21)

The second-order RLC low-pass filter designed in the pre-lab (R = 433 Ω, L = 4.7 mH, C = 100 nF, f_c = 7400 Hz) was assembled on the breadboard and characterised from 100 Hz to 32 kHz.

Frequency (Hz)	Vin (mV)	Vout (mV)	Gain	Phase (°)
100	352	325	0.923	1.8
320	351	349	0.994	0.47
1000	350	343	0.980	14.8
3200	334	285	0.853	44.7
10000	319	121	0.379	105.2
32000	333	17.8	0.053	170.6

All results are based on experimentally measured data, not numerical or circuit simulations.

The measured gain response rolls off more gradually than the ideal LTspice simulation, and the −3 dB point appears close to the target of 7400 Hz (confirmed by LTspice at 7.409 kHz, −6.018 dB). The phase reaches approximately 170° at 32 kHz, approaching the theoretical −180° limit for a second-order system.

The discrepancy between the measured and simulated Bode plots is primarily attributable to the parasitic resistances of the inductor and capacitor. In the LTspice ideal model, both components are treated as lossless; in practice, even a small parasitic resistance significantly alters the damping ratio and hence the shape of the frequency response, particularly near the corner frequency.